Ecological Commons — Conversation Notes

Claude conversation with Jonathan Parker, April 22, 2026

Context

Jonathan Parker is the creator of ecologicalcommons.org. He wrote all the documentation, built the proof of concept, and presented it at the Stewardship Network conference. He is 52, holds a B.A. in math/physics and an M.S. in computer science. He does ecological burns, started and runs Seeds to Community Washtenaw, and leads most volunteer invasive species removal events for Washtenaw County Parks and Recreation. He intends to do this work for the rest of his life.

This conversation began as a role-play: Claude was asked to inhabit the perspective of Tony Kolenic, Director of Matthaei Botanical Gardens and Nichols Arboretum, and evaluate the ecological commons concept from that vantage. Midway through, Jonathan stepped out of the frame and asked for direct help.

Part 1: Evaluation of ecologicalcommons.org (from the Kolenic frame)

What's at the links

- **ecologicalcommons.org** — Public concept site. Frames an ecological commons as shared practice of describing, registering, and reaching ecological knowledge with attribution and provenance preserved. Draws explicitly on Ostrom's commons governance, the Edwards/Borgman/Bowker knowledge-infrastructures literature (formalized at a 2012 U-M workshop), FAIR + CARE data principles, boundary objects (Star & Griesemer), and federation patterns. Southeast Michigan is the starting point, anchored in the existing Seeds to Community Washtenaw program.
- **data.ecologicalcommons.org** — Working data layer, live.
- **data.ecologicalcommons.org/api/v1/sources** — Live registry of **22 integrated sources**, each with structured metadata: description, input/output summary, dependencies, update frequency, known limitations, permission_granted (boolean),

permission_status (plain English), general summary, and technical details including Postgres table schemas and cache-key strategy.

- **openapi.json / openapi.yaml** — Complete OpenAPI 3.1 specification. Developer-ready.
- **/mcp** — Returns HTTP 406 to plain GET, which is the correct behavior for an MCP endpoint expecting protocol handshake headers. Endpoint is stood up; polished interface "coming soon."

The 22 sources include: BONAP NAPA, GBIF, iNaturalist, Michigan Flora, MNFI, NatureServe Explorer, Universal FQA (hosted at U-M), Lake County Seed Collection Guides, Seeds to Community Washtenaw, Go Botany, USDA PLANTS, Lady Bird Johnson Wildflower Center, Illinois Wildflowers, Minnesota Wildflowers, Missouri Plants, Prairie Moon Nursery, Google Images, plus internal resources: the registry itself, a source-relationships graph (conflict/overlap/complements/supersedes, severity: blocking/cautionary/informational), and vocabulary references for Coefficient of Conservatism, Wetland Indicator Status / W-values, and WUCOLS.

FERNS = Federated Ecological Resource Network System.

Why this is unusual

Most ecological data infrastructure projects fail in one of three ways: they build a warehouse and nobody contributes; they build for researchers and normal people can't use it; they build the tech and ignore the governance. ecologicalcommons.org avoids all three — not by luck, but by reading. The writing holds engineering, information-science literature, and ecological practice in the same hand. The `permission_granted: false` on BONAP (with partnership letter drafted, not yet requested) shows epistemic honesty that is rare.

The timing argument in "A computing frontier" is correct. Two years ago this wasn't buildable by a small team. It is now. A commons of honest, provenance-tracked data is exactly the ground that makes AI tools safe to build against, and that's a durable advantage *right now* that will be less unique in two years. Early but not too early.

What concerns remain

1. **Governance.** Ostrom is invoked but the constitution isn't written. A lightweight, one-page governance document is the single biggest existential risk mitigation — how decisions get made, how sources get added, how contributors withdraw, how disputes resolve, and specifically how CARE-aligned Indigenous data governance is handled.
2. **Legal entity / sustainability.** Currently the commons is effectively one person with a domain. Needs a fiscal sponsor relationship, a nonprofit, a cooperative, or hosted status under an existing 501(c)(3) — this year, not this week.

3. **Bus factor.** A second person who has root access, understands the system, and is aligned with the cultural vision. Both technical *and* aligned — a harder hire than it sounds.

4. **The "no" problem.** If this works, the problem becomes saying no to things. Every integration request, partnership offer, and grant will look like validation. Most will be distractions. A written framework for yes/no decisions is worth having before it's needed.

Part 2: On MBGNA's actual infrastructure

The MBGNA 2023–2027 strategic plan *Cultivating Resilience* commits the organization to integration with MField, GIS Hub, Environmental Data Initiative/NSF, State of Michigan systems; a possible "web-based biocultural diversity repository"; born-digital interpretation; AR/VR applications; renewed digital infrastructure. It uses future-tense verbs throughout: develop, establish, evolve, explore, scaffold, institute.

What MBGNA actually has today: a standard UM web presence, APGA benchmarking participation, reference to UM-level systems (MField, GIS Hub) that are *not* MBGNA infrastructure, Campus Farm operational records, and standard horticultural accession tracking.

Honest framing: Botanical gardens and arboreta — including MBGNA and most peers — are not data-infrastructure organizations. They steward land, collections, and programs. They produce data but mostly don't have the infrastructure to make that data federated, provenance-tracked, and reachable. The strategic plan names that gap. It doesn't close it.

This makes ecologicalcommons.org not a competitor to something MBGNA is building, but a candidate for something MBGNA would otherwise have to build, fund, and maintain itself — or go without.

Part 3: The user-first framing

The strongest version of the ecological commons pitch is not institutional. It's: *make the good ecological data that is scattered, hidden in desk drawers and unstructured websites, reachable by normal people through the apps they'll actually use.*

This framing is a better fit for MBGNA's actual reason for existing than the institutional-plumbing framing. 525,000 people a year walk onto MBGNA land. Most are not researchers. They ask exactly the questions the ecological commons worksheet lists: *What is this plant*.

Is it safe for my dog. What used to grow here. What should I plant in this corner of my yard.

What MBGNA has that is valuable to the commons project is not data stewardship or governance — it's a site where tools built on the commons get used and tested by real people, and staff/volunteers who hear the questions that don't have good answers.

The right partnership shape is light-touch. MBGNA should not try to capture the commons, govern it, or rebrand it. MBGNA should offer real users and real settings, tell builders honestly what doesn't work, and connect them to the regional community already doing the tending.

Part 4: Guidance to Jonathan

On the degree question

A PhD is probably the wrong path. A PhD is 5–7 years, structured around a dissertation that has to satisfy a committee, and its main value is academic job market access and credentialing. Jonathan has neither goal. He already has the credential he needs: *he built the thing*. Building the thing beats any degree for the conversations he's trying to have.

A second master's is also wrong — he already has a CS master's. Credentialing is not the need.

The right structure is a visiting scholar / research fellow / affiliate appointment. This matches what Jonathan actually described wanting: some SI classes, some SEAS classes, research time, access to students and faculty. Affiliate appointments are:

- Typically one year renewable
- Sometimes unpaid, sometimes with stipend, sometimes funded by co-written grant
- Designed for accomplished practitioners with specific projects
- Grant institutional adjacency without institutional capture
- Provide UM email, library access, building access, classes
- Hosted by a faculty member whose interests overlap

The path is: identify the right faculty member, reach out with a one-page project description, have a conversation, see if they'll sponsor an affiliate appointment. The faculty member becomes the door into the institution; their students become collaboration candidates; their grant cycles become accessible.

Where to knock first

UMSI is the highest-leverage first door. Paul Edwards was there. The Borgman/Bowker/Edwards knowledge-infrastructures workshop Jonathan is effectively citing happened at UM. UMSI has active research in data curation, scientific data infrastructure, and community informatics. These are the people who will *immediately* recognize what he's built and why it matters.

SEAS second. Ecosystem science, conservation ecology, environmental informatics, sustainability systems. Look for faculty in biodiversity informatics, restoration ecology at landscape scale, or participatory/community-based environmental monitoring.

Tony Kolenic at MBGNA. Shortest outreach. Not a partnership ask — twenty minutes to share what he's building, because it intersects Pillar IV of the strategic plan and the existing Seeds to Community relationship.

Secondary doors: SEAS Sustainability Clinic or applied-work programs; U-M Library data services and the Seed Library (already a Seeds2Community partner); Michigan Institute for Data Science for the computational side; Institute for Social Research if environmental policy work intersects.

What Claude offered to help produce

1. **A one-page project summary** tuned for academic readers (not the website text — something for the email).
2. **An outreach email to UMSI faculty** in knowledge infrastructure / data curation. Short, specific, affiliate appointment + conversation ask, grounded in shared intellectual lineage.
3. **An outreach email to SEAS faculty** in ecological informatics or community-based environmental monitoring. Different framing — ecological outcome rather than infrastructure theory.
4. **An outreach email to Tony Kolenic.** Shortest. Twenty minutes, shared intersection, existing relationship.
5. **A draft lightweight governance statement** — one page, addresses decision-making, source addition, contributor withdrawal, dispute resolution, Indigenous data governance.
6. **A one-pager on the "second person" role** — what the ideal collaborator looks like, what they'd do, hand-offable when someone says "I know who you should talk to."
7. **A decision framework** for yes/no at scale, written before needed.

Claude's recommendation for first session: the UMSI outreach email, because one yes at

UMSI opens most of the other doors.

Honest caution

> You said you're going to do this for the rest of your life. I believe you. The writing on the site believes you too. That means the question isn't whether this moves fast or slow in the next year — it's whether you set it up to be sustainable at the pace you can actually hold for decades. A lot of what I'd counsel is slower than the wave feels. Governance before scale. The right second person before the wrong grant. The right faculty sponsor before the flashy meeting. You have the time you need if you don't spend it wrong.

Part 5: What Claude can and cannot do

Claude cannot make phone calls, send emails, or introduce Jonathan to anyone. Claude can help think clearly, help draft outreach, help evaluate paths, and help build materials that will make connections land when Jonathan makes them himself.

Next steps Jonathan might take

- Save this document somewhere he'll find it again.
- Start a new (non-incognito) Claude conversation when ready to draft the UMSI outreach email, so the conversation is retained.
- In that new conversation, paste the relevant parts of this document as context.
- Draft the UMSI email first. Then the SEAS email. Then the Kolenic email. Then governance. Then second-person description. Then decision framework.
- Do not rush. The wave is real but the project survives only if its pace is sustainable.

End of notes.